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CSCE 3530 Section 401

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Project Work

1. Project Overview:
   This project involves designing, building, and testing a working enterprise-to-WAN network in Cisco Packet Tracer. The network connects a private Local Area Network (LAN) to the isolated external Wide Area Network (WAN) public web server segment. This project allows for the generation of traffic and analysis of packet changes through the network to study data link, network, transport, and upper-layer interactions and behaviors.

2. Objectives of the Project:
   Construct a functional topology that connects a local PC0 to the destination Server0 through an intermediate Switch0 and Router0

   Ensure the local host is dynamically configured using DHCP and DNS services and the Internet access is secured via NAT and encrypted using HTTPS.

   Capture and analyze the packet content and routing information to examine the differences and similarities in the packet content and routing information.

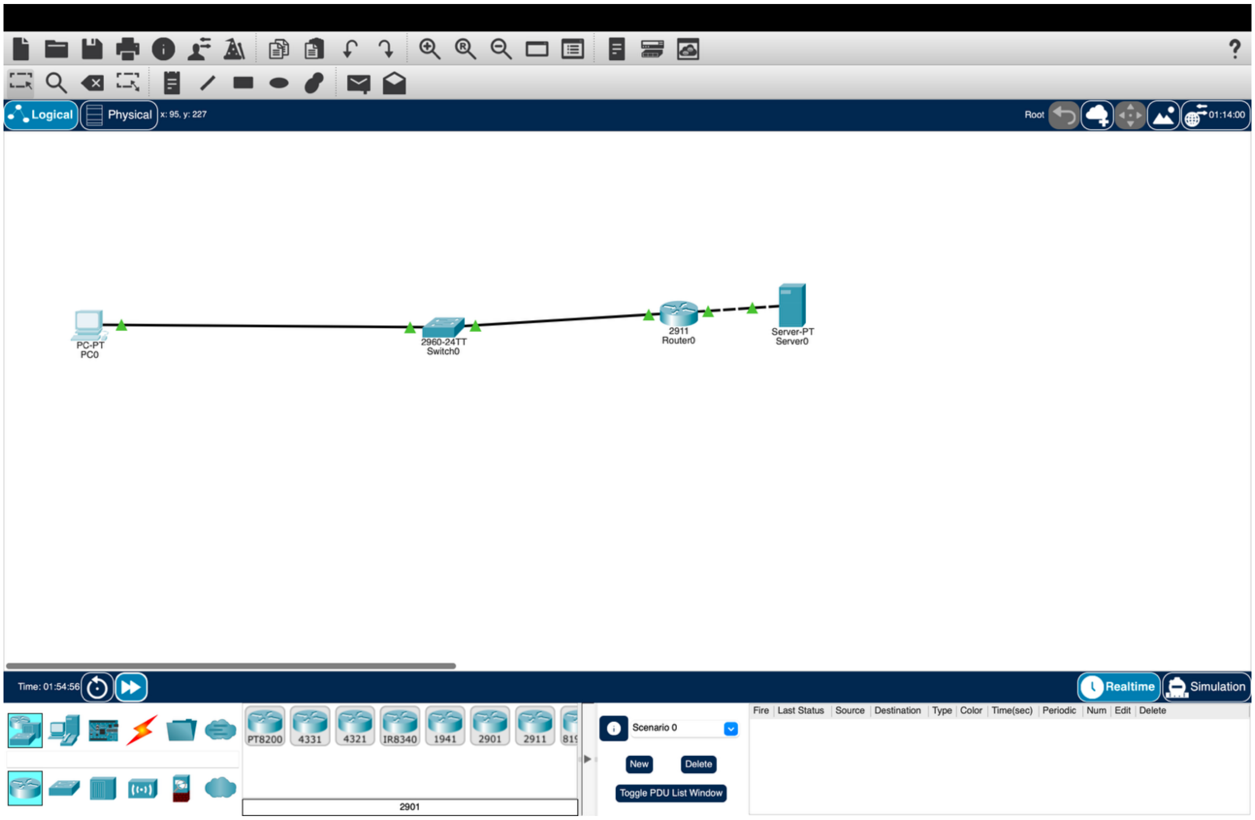
   Use the packet capture tool to view the packet content and routing information and describe the observed behavior in the packet content and routing information.

3. Network Design Explanation:
   The network is divided into two Collision/Broadcast Domains separated by a Layer 3 gateway (Router0).

The Private LAN Corridor uses a standard private IPv4 space classful designation to define a /24 (192.168.1.0/24) subnet that serves the clients local to this domain via a Layer 2 Switch (Switch0).

The Public WAN Link uses a public test block (203.0.113.0/24) which acts as a stand-in for the production Internet-facing edge with a data center housing a public Web Server (Server0).

Device	Interface	IP Address	Subnet Mask	Default Gateway	Purpose
Router 0	GigabitEthernet0/0	192.168.1.1	255.255.255.0	N/A	LAN Default Gateway (NAT Inside)
Router 0	GigabitEthernet0/1	203.0.113.1	255.255.255.0	N/A	WAN Edge Interface (NAT Outside)
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1	End-User Client Host (DHCP Assigned)
Server 0	FastEthernet0	203.0.113.10	255.255.255.0	203.0.113.1	Target Web & DNS Server Host



4. Description of Networking Protocols Used:
   DHCP (Dynamic Host Configuration Protocol): Used for assigning IP addresses, subnet masks, default gateway, and DNS server addresses to local hosts. Eliminates the need for static IP address configurations.

   DNS (Domain Name System): Converts human-readable strings like “www.project.local” to associated layer-3 routable destinations through a UDP Port 53 query.

   ARP (Address Resolution Protocol): Resolves known IP destination addresses to a physical data-link layer hardware identifier, allowing headers to be built for frame-based communications.

   ICMP (Internet Control Message Protocol): Family of message types used for pings and traceroute operations.

TCP & UDP: Transport-layer protocols responsible for defining how information is transferred. UDP is used for connection-less services like DNS, while TCP manages connection sequencing for services like HTTP and HTTPS.

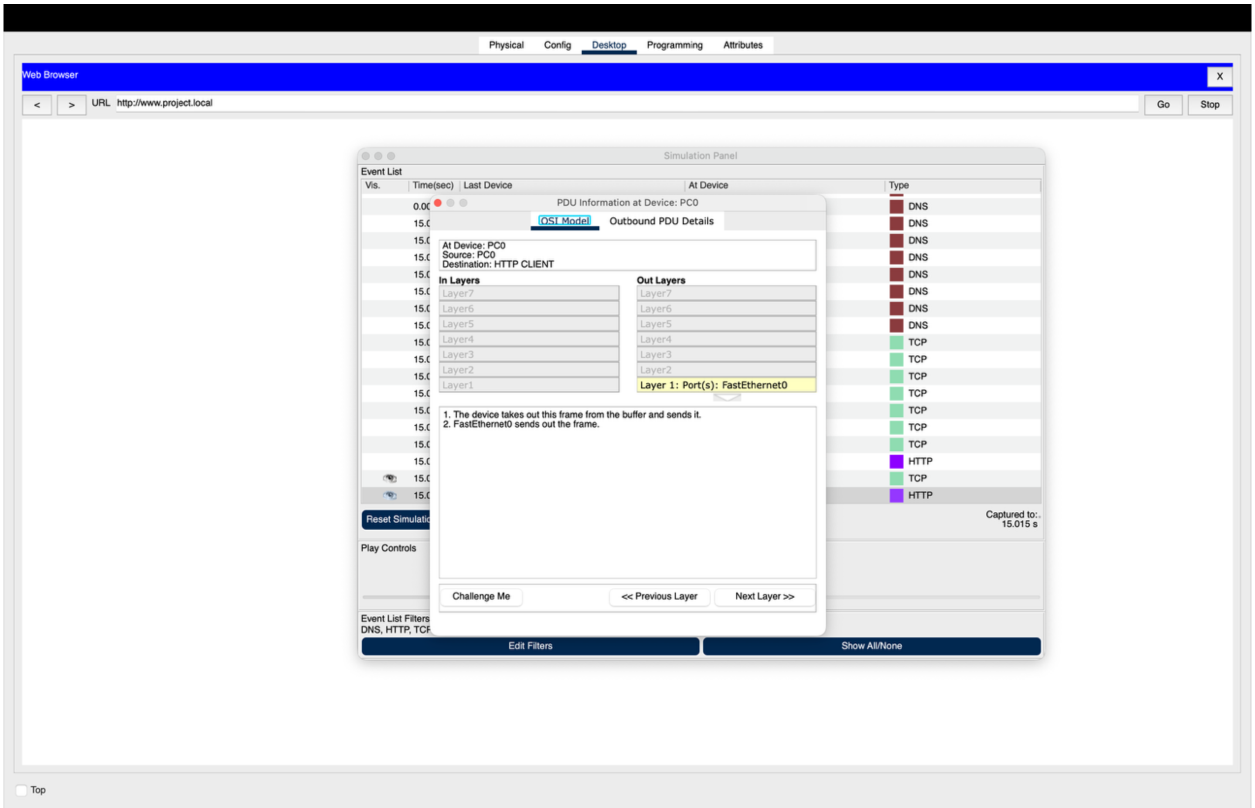
HTTP/HTTPS: Application-layer protocol for managing web data transfer. HTTPS utilizes an SSL/TLS encrypted TCP Port 443 session to ensure data privacy between the client and web server.

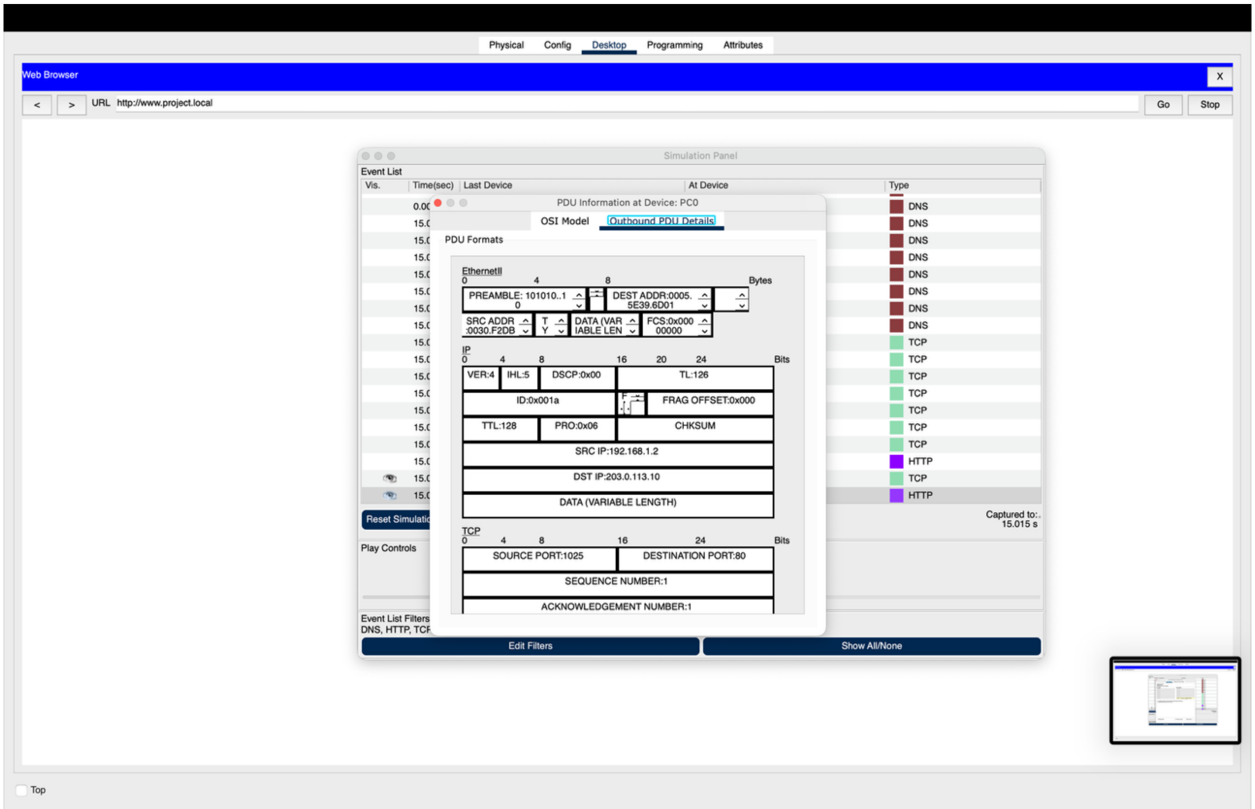
NAT (Network Address Translation – Overload/PAT): Used for allowing private RFC 1918 network IP addresses to connect to the Internet while hiding them behind a single gateway address. This conserves IPv4 addresses and provides a degree of anonymity for local area network (LAN) architecture information from external sources.

5.Device Configurations:
The gateway router was fully provisioned via the Cisco IOS Command Line Interface to host the native DHCP management pool, define layer-3 interface parameters, establish access control criteria, and enforce Port Address Translation

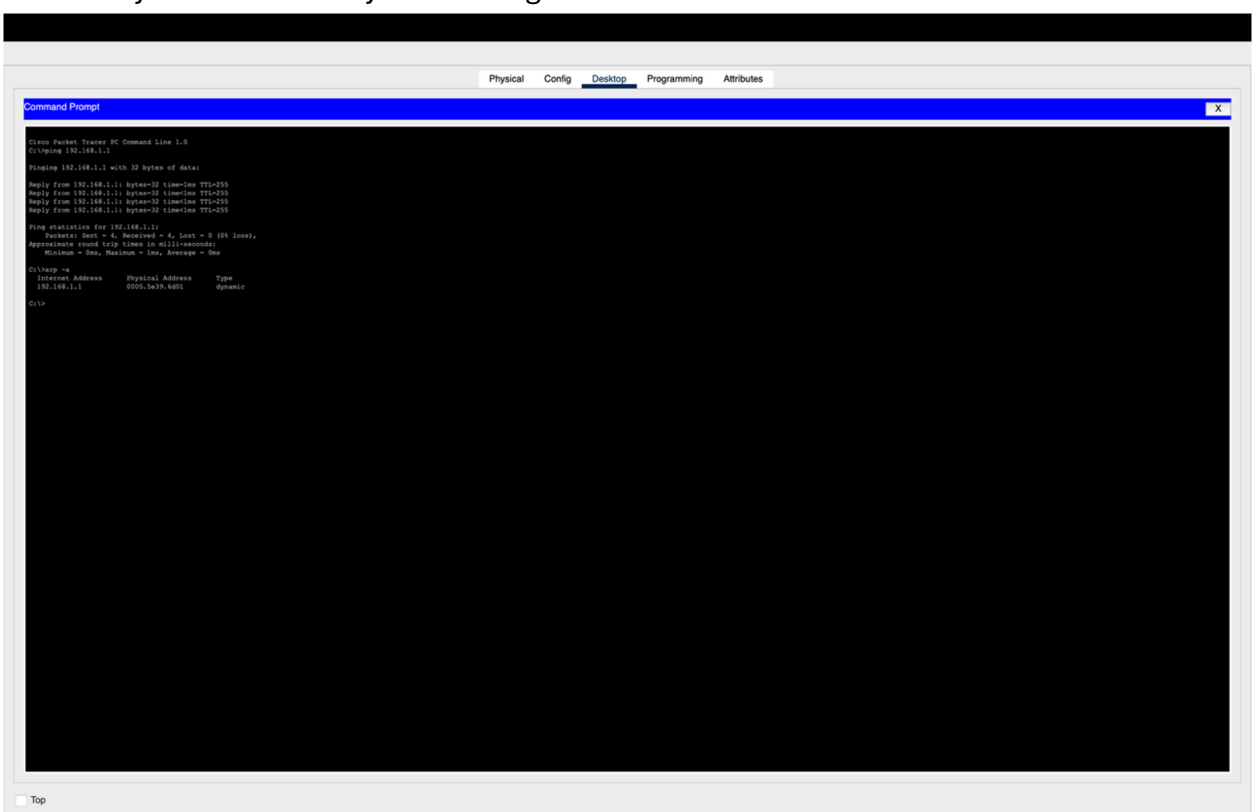
6. Network Testing and Results:

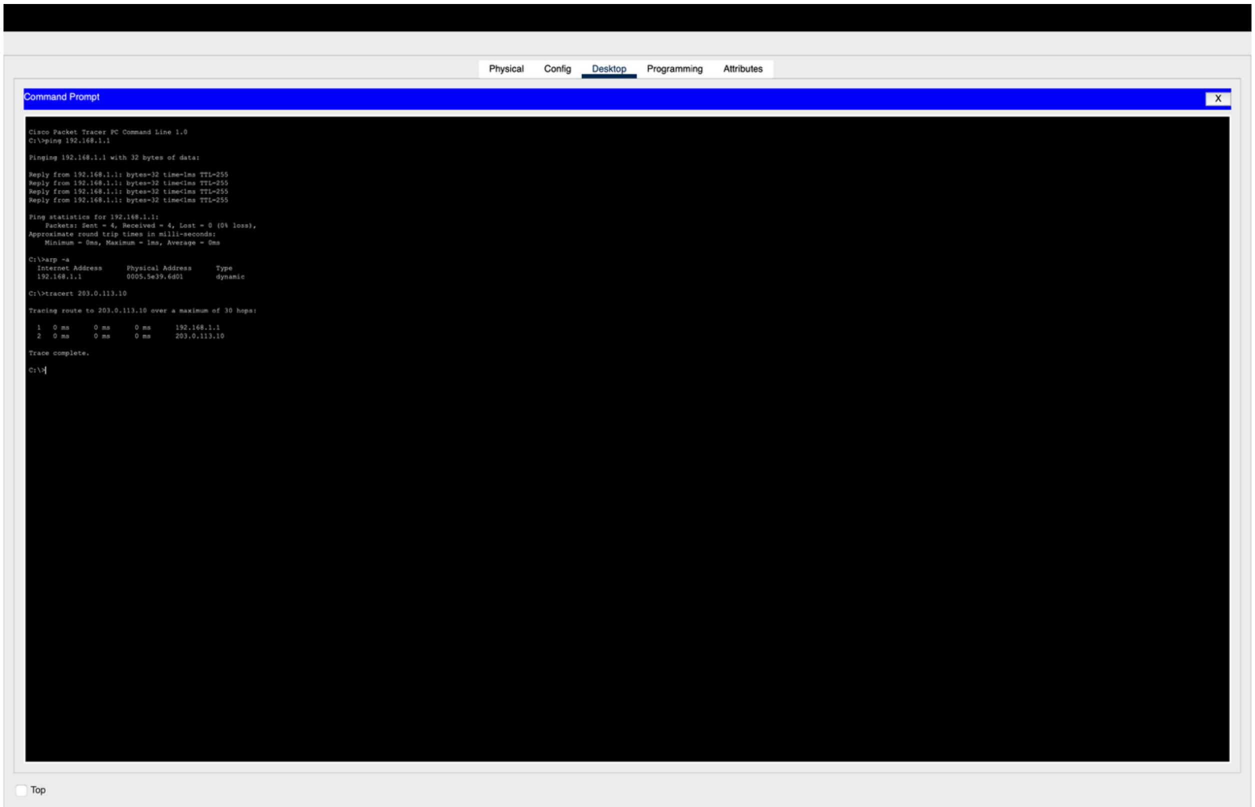
Part A: Dynamic Packet Captures


Physical
Config
Desktop
Programming
Attributes
Web Browser
X
URL http://www.project.local
Go
Stop
Simulation Panel
Event List
Vis.
Time(sec)
Last Device
At Device
Type
PDU Information at Device: PC0
OSI Model
Outbound PDU Details
At Device: PC0
Source: PC0
Destination: HTTP CLIENT
In Layers
Out Layers
Layer 1: Port(s): FastEthernet0
1. The device takes out this frame from the buffer and sends it.
2. FastEthernet0 sends out the frame.
Challenge Me
<< Previous Layer
Next Layer >>
Captured to:
15.015 s
Play Controls
Event List Filters
Edit Filters
Show All/None
Top

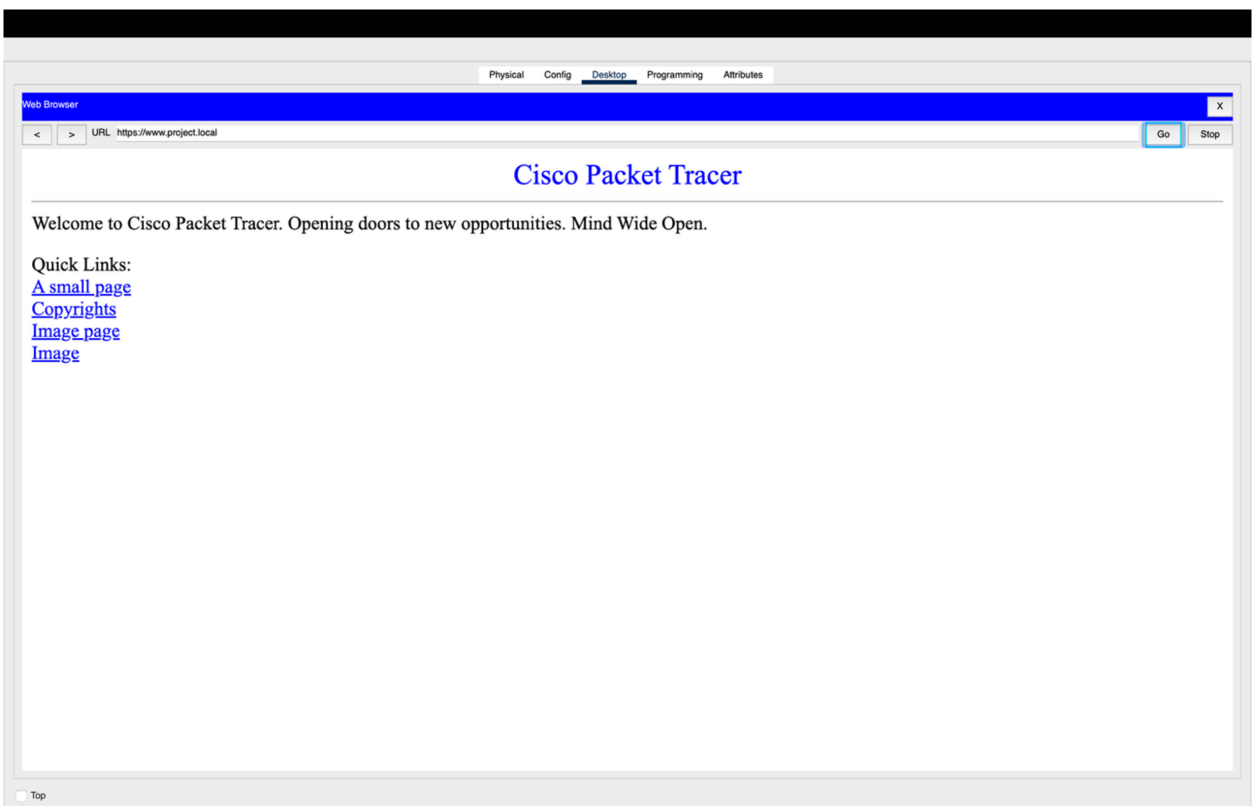


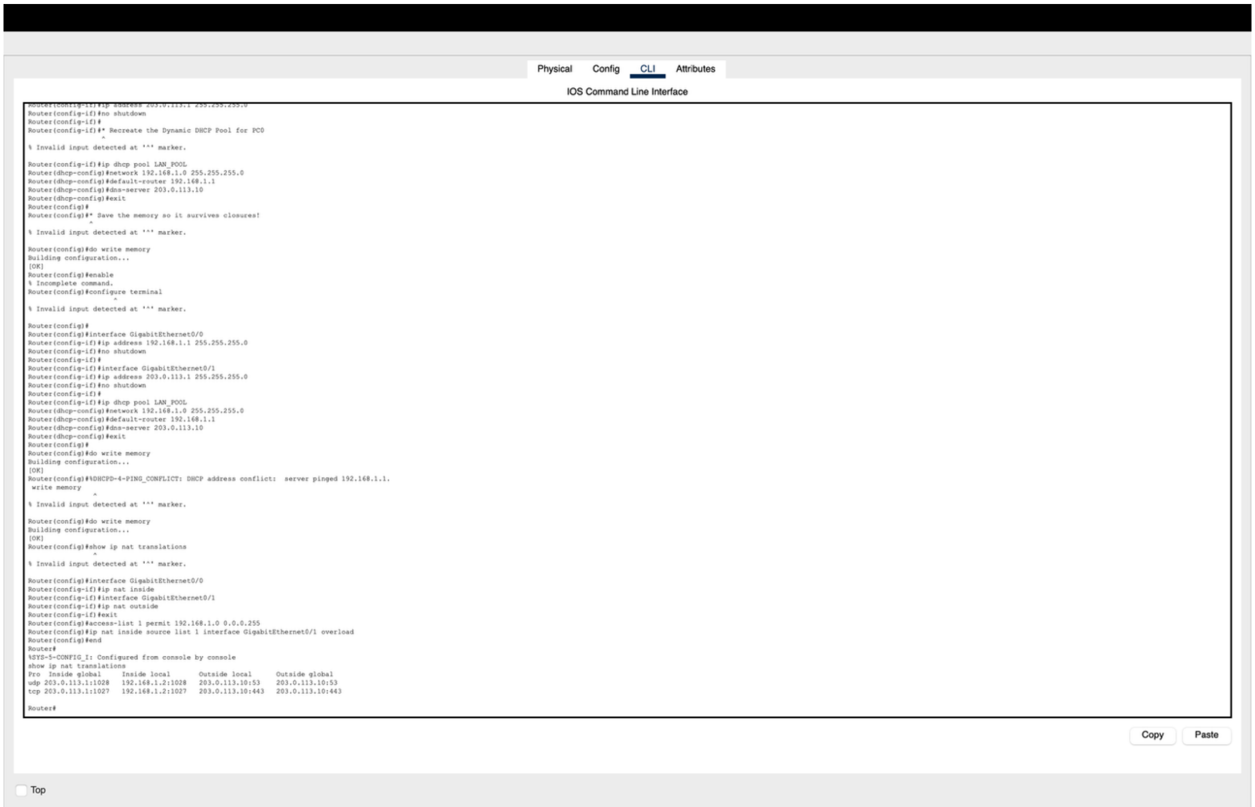
Part B: Layer 3 Connectivity and Testing:





Part C(Application Layer Validation):





7. Challenges Faced and Solutions:

The initial show ip nat translations lookups returned an empty dataset, leading to un-translated public routing failures. It occurred to me that access lists had to be reviewed so that local tracking masks matched exactly as they were. It was concluded that active traffic had to originate from the host browser with an interface direction vector in place to populate the translation table.

8. Conclusion:

   This project was able to model an integrated, reliable network lifecycle from the definition of physical links at the bottom to the stateful application layer on top. The examination of automatic addressing (DHCP), identity resolution (DNS), route path discovery (ICMP), cryptographic transport protection (HTTPS), and border security address mapping (NAT) all performed according to specification. The produced results document a fully formed and operational system environment that is ready for production deployment.